

Solution

WORK-01

Class 09 - Science

1. (a) sound will be louder but pitch will not be differ

Explanation: Pitch will not change because loudness depends on amplitude. By hitting it hard it will make its amplitude higher. But the frequency will be same. And the point is to be noted is that the loudness depends on the amplitude not the frequency.

2.

(c) 10 m

Explanation: Given $T = 0.1$ sec

$$V = 50 \text{ m/s}$$

$$V = v \lambda = \frac{\lambda}{t}$$

$$50 = \frac{\lambda}{0.1}$$

$$\lambda = 5\text{m}$$

The distance between two consecutive crests is $= 2 \times 5 = 10 \text{ m}$

3.

(c) So, that echo effect may be minimized.

Explanation: The roof and walls of the auditorium are generally covered with sound-absorbent materials to reduce reverberation. These materials reduce the formation of echoes by absorbing sound waves.

4.

(c) all of these

Explanation: Pulse is a short duration disturbance. Pulse does not repeat. Pulse is formed in small portion and Pulse disturbance is momentary.

5.

(d) Speed is comparatively higher

Explanation:

- Mechanical Waves require a medium, in order to transport their energy from one place to another.
- Mechanical Waves have two types of waves: Transverse and Longitudinal.
- Mechanical Wave is a wave that is not capable of transmitting its energy through empty space.
- Speed is comparatively higher is not a character of mechanical waves.

6. (a) A large size, the opaque reflecting surface

Explanation: A good reflector of sound and not light. Sound waves are much longer than light waves so they require a much larger area of reflection.

7.

(d) quality

Explanation: Quality is the sensation received by the ear by which, we are able to differentiate two sounds (even if they are of the same pitch and loudness). The quality actually depends on the shape of the waveform and the number of overtones.

8.

(c) energy

Explanation: 'Wave' is a common term for a number of different ways in which energy is transferred: In electromagnetic waves, energy is transferred through vibrations of electric and magnetic fields. In sound waves, energy is transferred through the vibration of air particles or particles of a solid through which the sound travels.

9.

(d) 20 Hz - 20,000 Hz

Explanation: The frequency range of human hearing is called an audible range of sound. The human ear is normally sensitive to only those vibrations whose frequency lies between 20 Hz to 20000 Hz.

10.

(c) supersonic speed and pressure variation

Explanation: A sonic boom is a sound associated with the shock waves created whenever an object traveling through the air travels faster than the speed of sound. Sonic boom created by an aerospace vehicle when the vehicle is flown at supersonic speed. The pressure from sonic booms caused by aircraft often is a few pounds per square foot.

11. (a) $v_w > v_a$

Explanation: Sounds in water and sounds in the air that has the same pressures have very different intensities because the density of water is much greater than the density of air and because the speed of sound in water is much greater than the speed of sound in air.

12. (b) amplitude

Explanation: The loudness or softness of a sound is determined basically by its amplitude. So for a loud sound it must have higher amplitude.

13. (b) ultrasonic

Explanation: Sound of frequencies above 20,000 Hz is known as ultrasonic which are produced by bats and dolphins.

14. (b) $f_i < f_a < f_u$

Explanation: Infrasound frequency is lesser than 20 Hz .
Audible sound frequency lies between 20 Hz to 20000 Hz.
Ultrasound wave frequency is more than 20000 Hz .

15. (d) persistence

Explanation: The persistence of sound due to the multiple reflections of sound in an auditorium is called reverberation. It can be reduced by wall proofing and roof proofing the auditorium with sound-absorbent materials like compressed fiberboard. The seat should have sound-absorbing properties.

16. (d) (a) - (iii), (b) - (i), (c) - (iv), (d) - (ii)

Explanation: (a) - (iii), (b) - (i), (c) - (iv), (d) - (ii)

17. (a) Vacuum

Explanation: Higher density and incompressibility brings higher speed for sound.

18. (c) Longitudinal wave

Explanation: Sound is a longitudinal wave.

19. (b) 10^6 times

Explanation: The speed of sound in air is 344 m/s. The speed of light in air is about 300,000,000 meters per second = 300×10^6 m/s.

20. (d) ultrasonic waves

Explanation: SONAR stands for Sound Navigation And Ranging. This is a device that is used to measure depth of sea bed, locate scraps, wrecks, submarines of enemies, etc. in the water by producing ultrasound. It is fitted over ships and submarines. Ultrasonic sound waves are produced by transmitter of SONAR and when these waves return back to the detector, after striking the object in the sea bed, they are analyzed with the help of computer. The shape and position of objects under sea and ocean is detected on the basis of speed and nature of reflected ultrasound waves.

21. (c) 55°

Explanation: The angle of reflection of sound is always equal to angle of incidence of sound.

$$\angle r = \angle i = 55^\circ$$

22. (c) rhinoceros

Explanation: The audible range of sound for human beings extends from about 20 Hz to 20,000 Hz. Sounds of frequencies below 20 Hz are called infrasonic sound or infrasound. Animals like dogs, elephants, rhinoceros, whales etc. produce and hear infrasonic sound.

23.

(d) vacuum

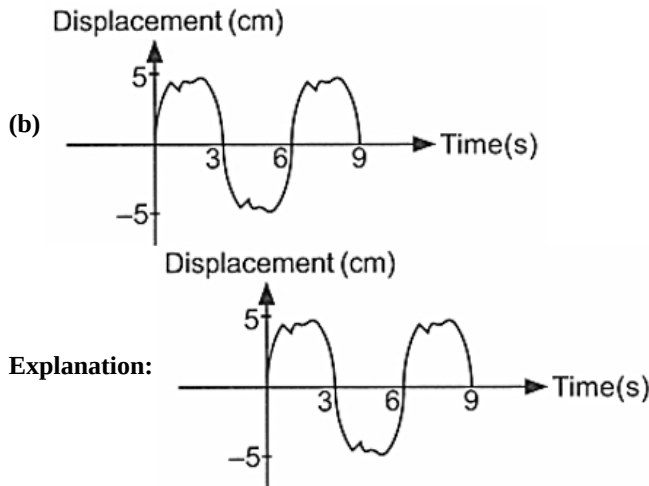
Explanation: Sound needs a material medium like solid, liquid, or gas to travel and be heard because the molecules of solid, liquid, and gases carry sound waves from one place to another. Sound cannot travel through the vacuum or empty space because the vacuum has no molecules which can vibrate and carry sound waves.

24.

(c) oscillates

Explanation: Oscillation will be perpendicular to the length or direction of energy transfer. Activity that carries energy from one place to another without actually moving any matter is a wave motion.

25.



26.

(b) by giving a jerk to slinky in vertically upward direction.

Explanation: A transverse wave is developed. A wave or pulse is generated in the slinky. While moving the pulse through the slinky, a particular position of the wave is noted and the timer is started.

27.

(c) 10 cm

Explanation: Wavelength = 2(distance between a compression and next rarefaction) = $2 \times 5 = 10\text{cm}$

28.

(c) note

Explanation: The sound that is produced when several frequencies are mixed is called a note. The unpleasant sounds around us are called noise. The sounds which are pleasant to hear are called all musical sound.

29. (a) Rhinoceros

Explanation: Rhinoceroses communicate with one another by using the infrasonic sound of frequency as low as 5 Hz.

30.

(d) Increases

Explanation: When compression takes place in the medium, the **density and pressure of the medium increase**. When rarefaction takes place in the medium, density, and pressure of the medium decrease.

31.

(b) 1.3 s

Explanation: Time taken by the sound produced by man to reflect back from first wall = 0.5 s.

Time taken by sound produced by man to reflect back from second wall = 0.8 s

A further echo is heard when sound reflected from first wall suffer one more reflection from second wall.

$\therefore$ The further echo is heard at a time of $(0.5 + 0.8)$ s or 1.3 s.

32.

(d) 0.1 m^{-1}

Explanation: Wave number = $\frac{1}{\lambda} = \frac{1}{10} = 0.1 \text{ m}^{-1}$

33.

(c) 0.25 cm

Explanation: Frequency, $\nu = 120 \text{ KHz} = 120 \times 1000 \text{ Hz}$

Initial velocity $u = 300 \text{ m/s}$

Wavelength $\lambda = \frac{u}{\nu} = \frac{300}{120 \times 1000} \text{ m} = \frac{300 \times 100}{120 \times 1000} \text{ cm} = \frac{1}{4} = 0.25 \text{ cm}$

34.

(b) different parameters and so cannot be related

Explanation: Ultrasonic waves are acoustic waves that are so high in frequency that humans can't hear them; however, infrasonic waves are sound waves that are lower in frequency than what humans can hear. A subsonic wave is a wave that is traveling slower than the speed of sound and supersonic wave travels faster. The terms ultrasonic, supersonic, and infrasonic mean different parameters and so cannot be related.

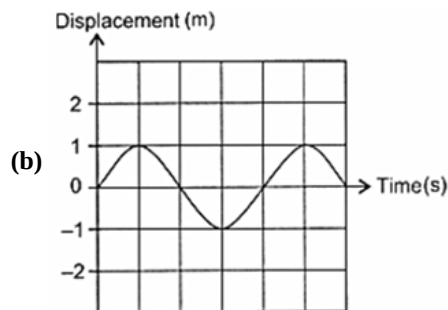
35.

(b) $1.4 \times 10^{-2} \text{ m}$

Explanation: $v = \nu \lambda \Rightarrow \lambda = \frac{v}{\nu} = \frac{340}{20,000} = 0.017 \text{ m} = 1.7 \times 10^{-2} \text{ m}$

$\therefore$ The only possible wavelength is $1.4 \times 10^{-2} \text{ m}$

36.



Explanation: The sound wave having less amplitude will be softer.

37.

(a) pressure

Explanation: A sonic boom is caused by the pressure change across the shock wave. The pressure from sonic booms caused by aircraft often is a few pounds per square foot.

38.

(b) reverberation time has to be decreased

Explanation: To reduce reverberation in big halls and auditoriums, the roofs and walls are covered by sound-absorbing materials like compressed fibre boards, rough plaster or draperies.

39.

(c) infrasound

Explanation: Earthquakes produces low frequency (i.e., 5 Hz) infrasound before the main shock waves begin which possibly alert the animals and they get disturbed.

40.

(b) Multiple reflections of sound

Explanation: Multiple reflections of sound is the reflection of sound back and forth several times between reflecting surfaces. Due to multiple reflections, the sound waves get added up, this causes an increase in the amplitude of the sound. This is the principle used in the stethoscope and concert halls.

41.

(d) 0.345 m

Explanation: $\lambda = \frac{v}{f} = \frac{345}{1000} = 0.345 \text{ m}$

42.

(c) Be the same as that of the source

Explanation: Frequency of the sound waves is independent of the medium. It will be same in water as well as in air.

43. **(b)** (i) and (iii) only
Explanation: (i) and (iii) only
44. **(a)** reflection
Explanation: The stethoscope is a medical diagnostic instrument based on multiple reflections of sound waves. Megaphone is a horn-shaped tube which is working on reflection of sound waves.
45. **(d)** after reflection, sound waves spread evenly across the width of the hall.
Explanation: A curved soundboard may be placed behind the stage so that sound after reflecting from the soundboard spreads evenly across the width of the hall. And regular reflection takes place.
46. **(a)** Long, soft and flexible
Explanation: More elastic and lengthier the spring will be, the motion of a pulse can be studied better.
47. **(b)** the angle of reflection is always equal to the angle of incidence
Explanation: Laws of reflection:
 a. The angle of incidence is equal to the angle of reflection.
 b. The incident ray, the reflected ray and normal at the point of incidence all lie in the same plane.
48. **(a)** $f_1 > f_2$
Explanation: The audible range of sound for human beings extends from about 20 Hz to 20,000 Hz. Sound whose frequency is more than 20000 Hz is called ultrasonic sound.
 The penetrating power of ultrasonic waves f_1
 And other audible sounds = f_2
 $f_1 > f_2$
49. **(d)** All of these
Explanation: Sound is produced due to the vibration of an object. The motion of objects causes vibration. Vibration is a kind of rapid to and fro motion of an object about a central position.
 Example:
 ■ It can be produced by clapping of palms, beating a table, or drum.
 ■ Sound is produced by a guitar and flute because of the vibration of the guitar's string and vibration of the air column of flute respectively.
 ■ The sound of the human voice is produced due to the vibration of the vocal cords.
 ■ A stretched rubber band, when plucked vibrates and produces sound.
50. **(a)** 65°
Explanation: $\angle r = \angle i$
 So $2i = 130^\circ$
 $\Rightarrow i = \frac{130}{2} = 65^\circ$
51. **(d)** Statement 1 is true but statement 2 is false.
Explanation: Sound energy gets partially absorbed by objects with soft and loose texture.
52. **(d)** 300 m
Explanation: Given $T = 0.1$ sec
 $V = 3000$ m/s
 $V = v \lambda = \lambda/t$
 $3000 = \lambda/0.1$
 $\lambda = 300$ m

53.

(d) So that time taken by pulse to move from one end to string to other may be accurately determined.

Explanation: If any twists and bends in string then time taken by pulse to move from one end to other may not be accurate.

54.

(b) intensity falls

Explanation: As the air in the jar is being pumped out, the sound becomes fainter, although it is the same current that is passing through the bell. On reducing the air pressure further, the intensity of the sound reduces and stops.

55.

(d) $v_c > v_d$

Explanation: On a hot summer day, the Temperature of Chennai is greater than in Delhi.

Molecules at higher temperatures have more energy, thus they can vibrate faster. Since the molecules vibrate faster, sound waves can travel more quickly.

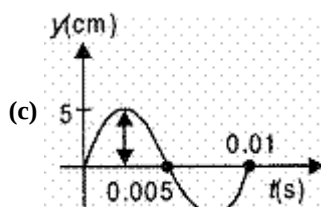
$$V_c > V_d$$

56.

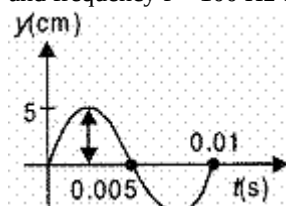
(d) Greater than 20,000 Hz

Explanation: Sound whose frequency is more than 20000 Hz is called ultrasonic sound. Animals like dolphins, bats, rats, porpoises, etc. produce and hear an ultrasonic sound.

57.



Explanation: In this graph, amplitude represents on y-axis and frequency on the x-axis. The wave having an amplitude of 5 cm and frequency $f = 100$ Hz can be best represented by a given graph.



58.

(b) 15 m

Explanation: The total distance covered by the sound from the point of generation to the reflecting surface and back should be at least $(300 \text{ m/s}) \times 0.1 \text{ s} = 30 \text{ m}$. Thus, for hearing distinct echoes, the minimum distance of the obstacle from the source of sound must be half of this distance. i.e., $30/2 = 15 \text{ m}$.

59.

(b) $\angle i = \angle r$

Explanation: Angle of incidence is equal to the angle of reflection.

Sound and light obey the laws of reflection $\angle r = \angle i$.

60.

(d) absence of medium

Explanation: Sound needs medium to travel. As there is no medium in the space, it can not travel from space to earth.

61.

(a) 5.36 s

Explanation: Initial velocity, $u = 0$ (body at rest)

Let t_1 be the time taken by the stone to reach the pond.

For a freely falling body

$$h = ut + \frac{1}{2}gt^2 \Rightarrow 125 = 0 \times t_1 + \frac{1}{2}10 \times t_1^2$$

$$t_1^2 = 25 \Rightarrow t_1 = 5 \text{ s}$$

Let the time taken by sound to reach the top = t_2

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}} \text{ or } 340 = \frac{125}{t_2} \text{ or } t_2 = \frac{125}{340} = 0.36 \text{ s}$$

Time after which splash will be heard at the top

$$= t_1 + t_2 = 5 + 0.36 = 5.36 \text{ s}$$

62.

(c) 0.5

Explanation: Speed of sound, $v_s = 330 \text{ m s}^{-1}$

Frequency of sound, $u = 660 \text{ Hz}$

$$\therefore \text{Wavelength of sound, } \lambda = \frac{v_s}{v} = \frac{330}{660} = \frac{1}{2} = 0.5 \text{ m}$$

63.

(d) Length of pipes should be sufficiently long

Explanation: Energy may be lost. To avoid the diffraction of sound and make its flow unidirectional long tube is used. The small diameter of the tube allows the selected sound wave to travel clearly.

64. **(a)** 803.8 m

Explanation: Time taken = 1.05 s

Speed of sound in salt water = 1531 m s^{-1}

Distance travelled by pulse from submarine to the cliff and back to submarine,

$$2 \times \text{Distance} = 1531 \times 1.05$$

$\therefore$ Distance between cliff and submarine,

$$S = \frac{1531 \times 1.05}{2} = 803.8 \text{ m}$$

65.

(b) Ultrasonics

Explanation: The sound of frequency more than 20,000Hz is called ultrasonics. Ultrasonic sounds have a frequency above the upper limit of human hearing.

66.

(b) $5 \times 10^{14} \text{ Hz}$

Explanation: The wavelength (λ) times frequency (v) equals the speed of light (c), which is $3.00 \times 10^8 \text{ m/s}$ to three significant figures.

$$c = \lambda \cdot v$$

$$v = c / \lambda$$

Frequency is measured in Hertz (Hz), which is /s or s^{-1} .

Convert 600 nm to meters since the unit for the speed of light is m/s.

$$1 \text{ nm} = 1 \times 10^{-9} \text{ m}$$

$$600 \text{ nm} \times \frac{1 \times 10^{-9} \text{ m}}{1 \text{ nm}} = 6 \times 10^{-7} \text{ m}$$

$$V = \frac{c}{\lambda}$$

$$V = \frac{3 \times 10^8 \text{ m/s}}{6 \times 10^{-7} \text{ m}} = 0.5 \times 10^{15} = 5 \times 10^{14} \text{ Hz}$$

67.

(c) The frequency and pitch of X is less than that of Y of X is less than y.

Explanation: The frequency of wave X is lower than wave Y, hence its pitch is lower than wave Y too. The pitch of sound depends on the frequency.

68. **(a)** axis of tube

Explanation: In the experiment of verification of reflection of sound, the incident sound is directed along because if it is normal to the axis of the tube then it will not form a proper angle. The angle between the incident ray and reflected ray may vary but the angle between the incident ray and normal ray have to be equal to the angle between the normal ray and the reflected ray.

69.

(c) 1 m, 114 m and 3 Hz

Explanation: Here, amplitude of sound wave = 1 m (given)

$$\text{Wavelength of sound wave, } \lambda = \frac{v_s}{v} \dots (i)$$

In 1 second, number of oscillations = $\frac{360}{2 \times 60} = 3$

∴ Frequency of sound wave, $\nu = 3 \text{ Hz}$

Put the value of ν in (i),

∴ Wavelength of sound wave, $\lambda = \frac{342}{3} = 114 \text{ m}$

70.

(b) Dimensions of slinky

Explanation: Dimensions include length but the only area of the coil may be taken. The dimensions of slinky do not affect the speed of propagation.

71.

(d) hertz

Explanation: The SI unit of frequency is the hertz (Hz), named after the German physicist Heinrich Hertz; one hertz means that an event repeats once per second.

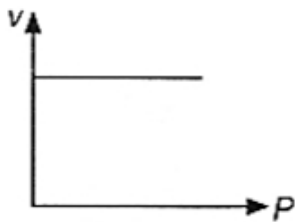
72.

(c) disturbance moves

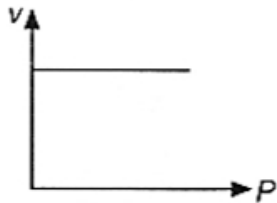
Explanation: Sound travels in air if disturbance moves. As during the propagation of sound waves, particles only vibrates in its own position and disturbances created by the vibration of particle moves from one place to other.

73.

(a)



Explanation:



74.

(b) of mixture of several frequencies

Explanation: Sound of mixture of several frequencies is called note. A note is pleasant to listen.

75.

(d) Ultrasound waves

Explanation: An echocardiogram uses high-frequency sound waves (ultrasound) to create a moving picture of your heart as it works.